

ENEE 132 — Basic Electronics I (Semiconductor Devices)

Lecture 7 - Special-Purpose Diodes

16 July 2026

- **Part 1 — The Varactor Diode**

- A reverse-biased diode used as a voltage-controlled capacitor, and how it tunes circuits

- **Part 2 — Optical Diodes**

- The LED in depth: how it makes light, materials, biasing, and applications
- High-intensity LEDs, OLEDs, quantum dots, and the photodiode (light detector)

- **Part 3 — Other Special-Purpose Diodes**

- Schottky, PIN, tunnel, laser, step-recovery, and current-regulator diodes

- **Part 4 — Wrap-Up**

- A quick-reference of which diode does what, and a look ahead to the transistor

Today's Learning Objectives

- **By the End of This Lecture, You Will Be Able To**

- Explain why a reverse-biased varactor acts as a voltage-controlled capacitor, and how it tunes a circuit
- Describe how an LED makes light (electroluminescence) and what sets its colour
- Size an LED's current-limiting resistor and describe its main applications
- Describe high-intensity LEDs, OLEDs, and quantum dots
- Explain how a photodiode converts light intensity into current
- Describe the Schottky, PIN, tunnel, laser, step-recovery, and current-regulator diodes and match each to its use

Part 1 — The Varactor Diode

- **Where We've Been**

- So far a diode has been a one-way valve — for rectifying, and (as a zener) for regulating

- **A Different Idea**

- Every diode has other behaviours we can exploit by design:
- Its junction capacitance (varactor), its ability to emit or detect light (LED, photodiode), its switching speed (Schottky, step-recovery), even a negative-resistance region (tunnel)
- For each device, note the one property it is built around and what it is used for

The Varactor — A Voltage-Controlled Capacitor

• A Diode Used as a Capacitor

- A varactor is a diode that always operates in reverse bias and is doped to maximize the capacitance of the depletion region
- The depletion region acts as a capacitor dielectric because of its nonconductive characteristic
- The p and n regions are conductive — they act as the two capacitor plates

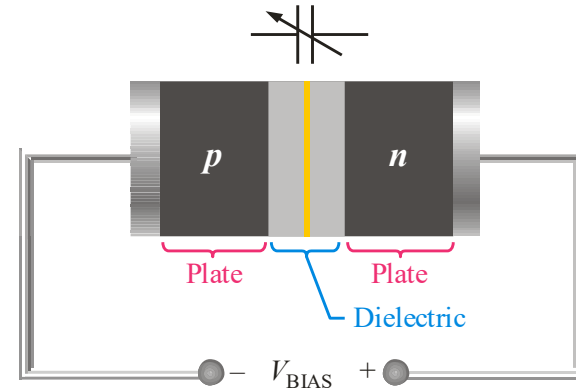


Fig 3.21 — The reverse-biased varactor acts as a variable capacitor

• The Capacitance

- The capacitance is set by plate area, the dielectric, and the plate separation (the depletion width)

$$C = \frac{A\epsilon}{d}$$



Fig 3.23 — Varactor diode symbol

The Varactor — Tuning Behaviour

- **Voltage Sets the Capacitance**

- More reverse voltage $\rightarrow$ wider depletion region $\rightarrow$ larger plate separation $\rightarrow$ less capacitance
- Less reverse voltage $\rightarrow$ narrower depletion $\rightarrow$ more capacitance
- Example device: CT falls from about 30 pF to under 4 pF as V_R goes from 1 V to 30 V

- **The Range**

- Nominal varactor capacitance runs from a few pF to several hundred pF (also called a varicap or tuning diode)

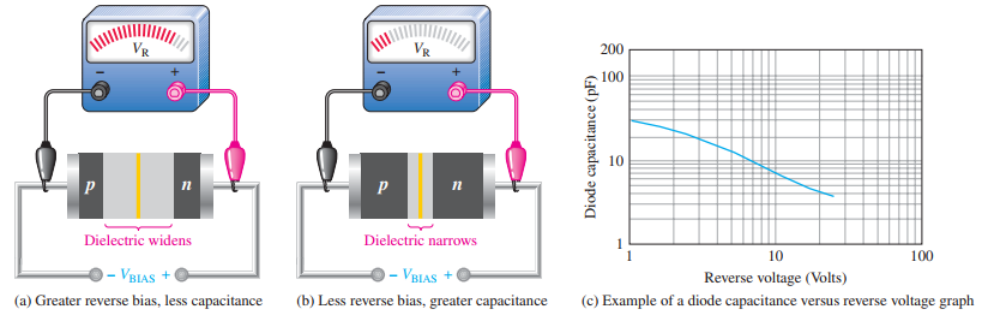


Fig 3.24 — Varactor capacitance varies with reverse voltage

Quick Check #1 — The Varactor Diode

- **Question 1**

- In what bias does a varactor always operate?

- **Question 2**

- Which part of the diode acts as the dielectric, and which parts act as the plates?

- **Question 3**

- As the reverse voltage increases, does the capacitance increase or decrease? Why?

Part 2 — Optical Diodes

The LED — Light Emitting Diode

- **LED**

- LED (Light Emitting Diode) emits light from recombination

- **How It Makes Light**

- Forward-biased: conduction-band electrons cross the junction and recombine with holes in the p region
- The energy released by each recombination comes out as a photon — this is electroluminescence

- **What Sets the Colour**

- The photon energy (and so the colour) is set by the band gap of the material
- The light is essentially monochromatic; some LEDs emit invisible infrared



Fig 3.28 — symbol for LED

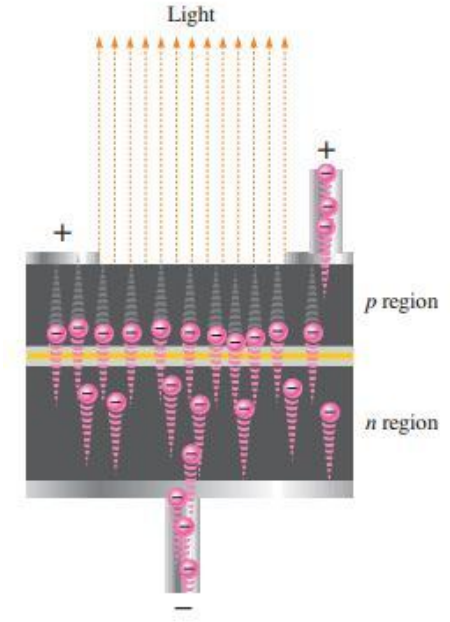


Fig 3.29 — Electroluminescence in a forward-biased LED

LED Semiconductor Materials, Colour & Wavelength

• Material Sets the Wavelength

- Gallium Arsenide (GaAs) for infrared radiation;
- Gallium Arsenide Phosphide (GaAsP) for red
- Gallium Phosphide (GaP) for brighter red, orange, and pale green
- Silicon carbide (SiC) and Gallium nitride (GaN) for blue;
- Indium Gallium Nitride (InGaN) for bright green/blue
- White LEDs: an ultrabright blue GaN LED coated with phosphor that re-emits part of the light as white

• Spectral Output

- Each LED emits over a narrow band; peaks: red ≈ 660 nm, yellow ≈ 590 nm, green ≈ 540 nm, blue ≈ 460 nm, infrared ≈ 940 nm

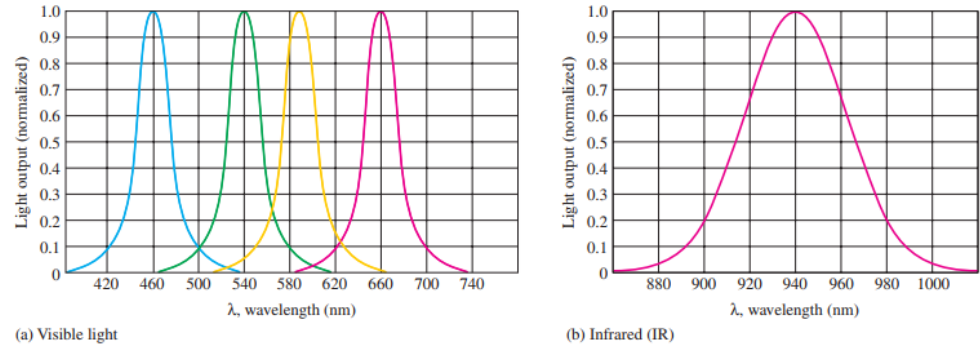


Fig 3.31 — Typical spectral output curves for LEDs

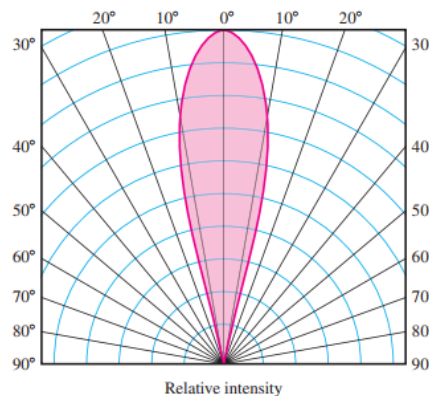
LED Radiation Patterns & Directionality

- **Directional by Nature**

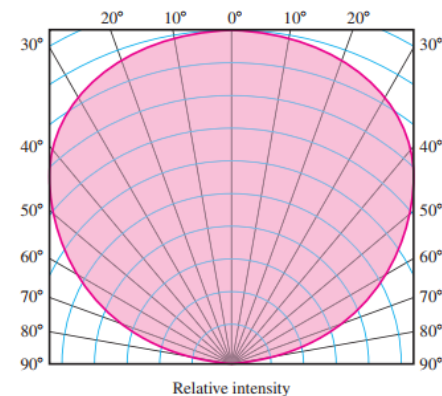
- Unlike a filament or fluorescent bulb, an LED emits in a preferred direction — generally perpendicular to the emitting surface
- The shape of the emitter, plus lenses and diffusion films, can steer the pattern

- **Two Common Patterns**

- Forward-directed narrow beam — used in small panel indicators
- Wide viewing angle — typical of super-bright LEDs
- Directionality is an advantage where light should reach only certain viewers (e.g. a traffic signal aimed at one lane)



(a) A narrow viewing angle LED



(b) A wide viewing angle LED

Fig 3.32 — Radiation patterns for two different LEDs

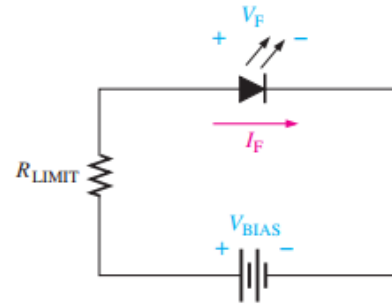
LED Biasing & Light Output

• Forward Voltage

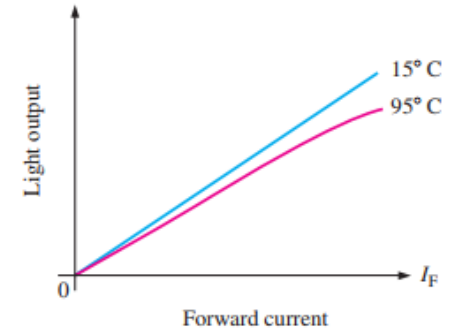
- Much higher than a silicon diode: typically $V_F = 1.2\text{ V}$ to 3.2 V (depends on colour/material)
- Reverse breakdown is low — typically only 3 V to 10 V

• Light vs. Current

- Light output is proportional to the forward current I_F — more current, more light
- Light intensity also falls as temperature rises; LEDs are directional emitters



(a) Forward-biased operation



(b) General light output versus forward current for two temperatures

Fig 3.30 — Basic operation of an LED

LED Forward Voltage by Colour

- **Colour Sets the Forward Voltage**

- Different-colour LEDs need different V_F ; red takes the least, and the requirement rises toward the blue end of the spectrum
- Red ≈ 2 V; blue ≈ 3 V to 4 V

- **But the Current Is Similar**

- Regardless of colour, most LEDs operate at about 20 mA to 30 mA
- The V-I curves show a sharp turn-on — once conducting, a small voltage change gives a large current change, so a series limiting resistor is essential

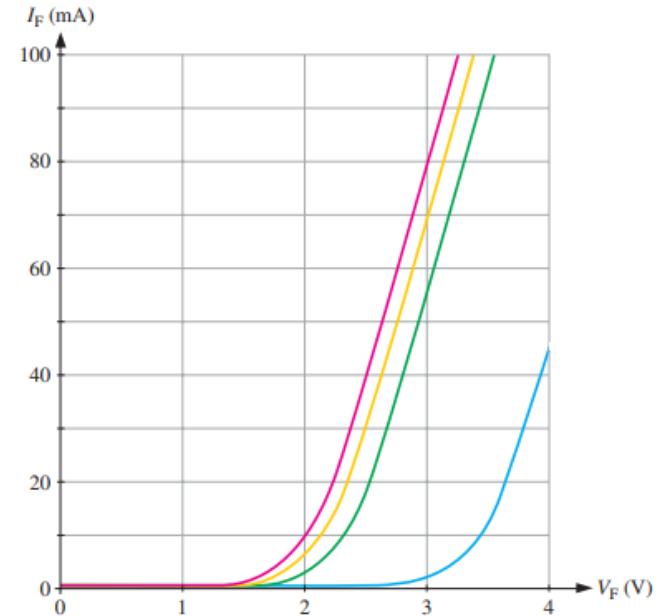


Fig 3.39 — V-I characteristic curves for visible-light LEDs

Example

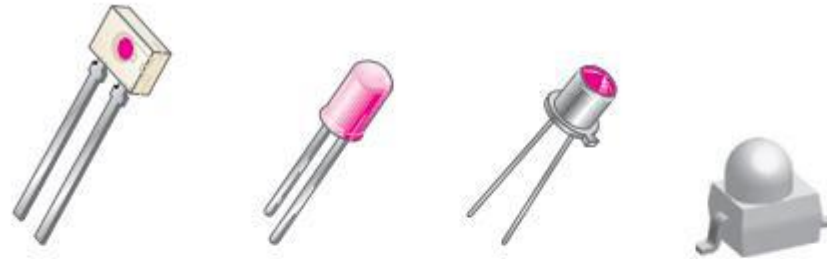
- **Question**

- A different bright LED drops 2.2 V at a maximum current of 20 mA. What series resistor is required to limit the current to 20 mA from a 5.0 V source

- **Solution**

$$R = \frac{V_s - V_{LED}}{I} = \frac{5.0 \text{ V} - 2.2 \text{ V}}{20 \text{ mA}} = 180 \Omega$$

Typical LEDs



(a) Typical small LEDs for indicators



Helion 12 V overhead light
with socket and module



120 V, 3.5 W screw base
for low-level illumination



120 V, 1 W small screw
base candelabra style



6 V, bayonet base
for flashlights, etc.

(b) Typical LEDs for lighting applications

Fig 3.33 — Typical LEDs

LED Applications — Indicators & Displays

• Indicators

- The most common use: panel indicator and status lamps on every kind of instrument and appliance

• Seven-Segment Displays

- Seven LED segments (plus a decimal point); lighting selected segments forms any digit 0–9
- Wired as common-anode or common-cathode, depending on the drive circuit
- Directional radiation patterns let the light be aimed — handy where only certain viewers should see it

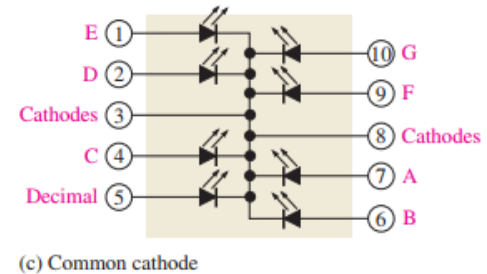
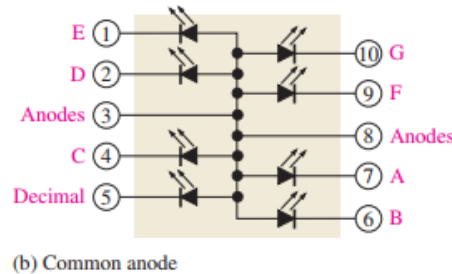
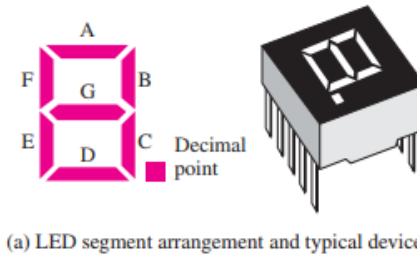


Fig 3.35 — The seven-segment LED display

Full-Colour LED Displays — The RGB Pixel

• Building a Pixel

- Full-colour screens group tiny high-intensity red, green, and blue LEDs into a single pixel
- A screen is thousands of these RGB pixels — the exact count depends on the screen and pixel size

• Mixing Any Colour

- Red, green, and blue are primaries; varying the forward current of each LED independently mixes any colour in the visible spectrum
- Some television screens add yellow as a fourth colour (RGBY)

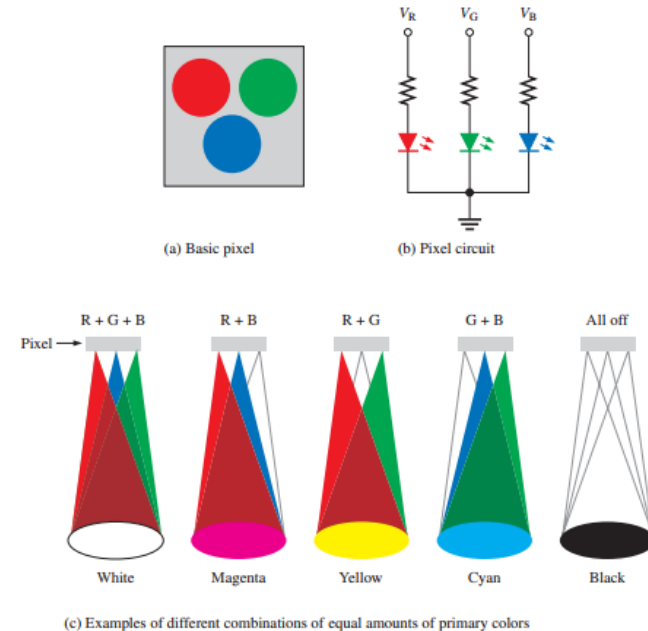


Fig 3.41 — The RGB pixel used in LED display screens

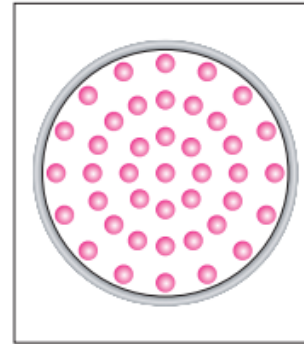
High-Intensity LEDs & LED Lighting

- **Much Brighter Output**

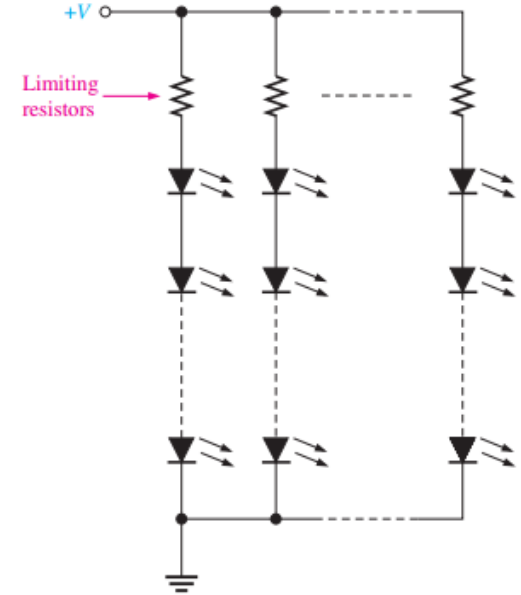
- Arrays of high-intensity LEDs now replace incandescent bulbs in many high-visibility roles

- **Where They're Used**

- Traffic signals — brighter, far longer life (years vs. months), and about 90% less energy than incandescent
- Automotive lighting, outdoor signs and displays, and general home and area lighting
- Efficiency is about 50–60 lumens per watt — roughly five times an incandescent bulb



(a) LED array



(b) Circuit

Fig 3.37 — A high-intensity LED traffic-light array

Worked Example — Designing an LED Array

• The Problem

- From the V-I graph a green LED drops about 2.5 V at $I_F = 20$ mA. Design a 12 V array of 60 LEDs using as few limiting resistors as possible.

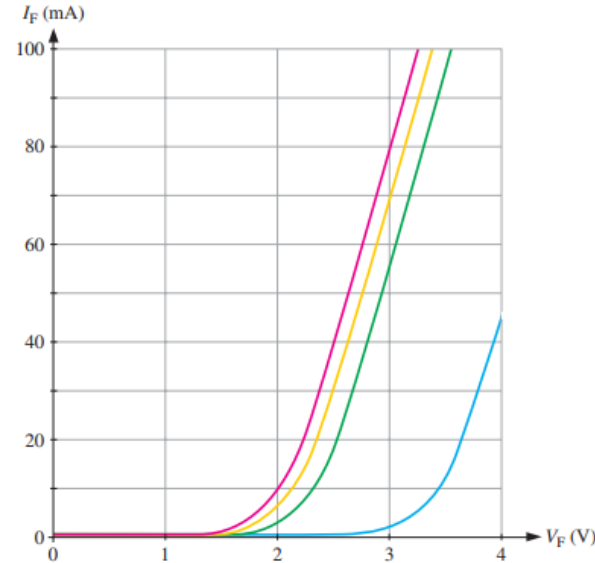


Fig 3.39 — V-I characteristic curves for visible light LEDs

Worked Example — Designing an LED Array Continued

- **Solution - Maximise LEDs per Branch**

- Put as many LEDs in series as the supply allows while leaving enough voltage across the resistor to hold the current steady: use 3 in series $\rightarrow 3 \times 2.5 \text{ V} = 7.5 \text{ V}$
- Resistor voltage: $V_R = 12 \text{ V} - 7.5 \text{ V} = 4.5 \text{ V}$
- Limiting resistor: $R = V_R / I_F = 4.5 \text{ V} / 20 \text{ mA} = 225 \Omega$

- **The Array**

- $60 \text{ LEDs} \div 3 \text{ per branch} = 20 \text{ parallel branches}$, each with three LEDs and one 225Ω resistor — 20 resistors instead of 60

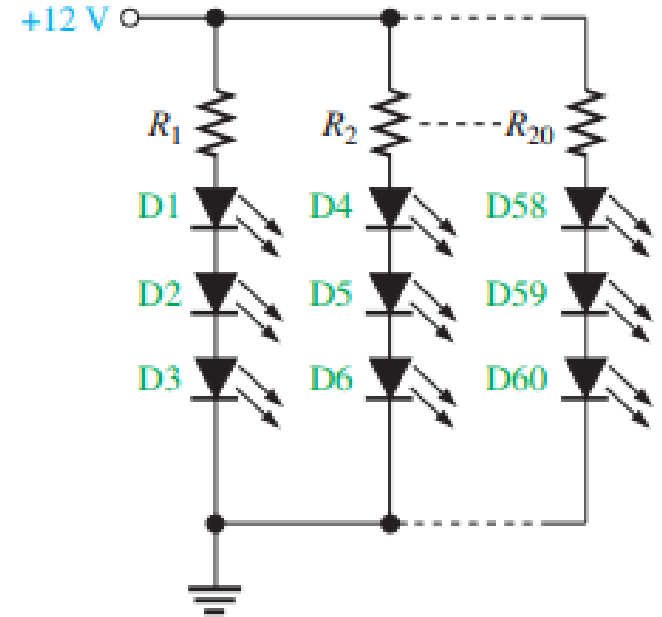


Fig 3.40 — The 12 V, 60-LED series-parallel array (20 branches of three)

The Organic LED (OLED)

- **Light from Organic Layers**

- Two or three layers of organic molecules or polymers emit light when current flows (electrophosphorescence)
- Electrons and holes meet near the layer junction and recombine, releasing light through a transparent contact

- **Why They Matter**

- Thin, flexible, and can be inkjet-printed onto large films — used for big TV screens and billboards
- If both contacts are transparent, light exits both sides — useful for heads-up displays

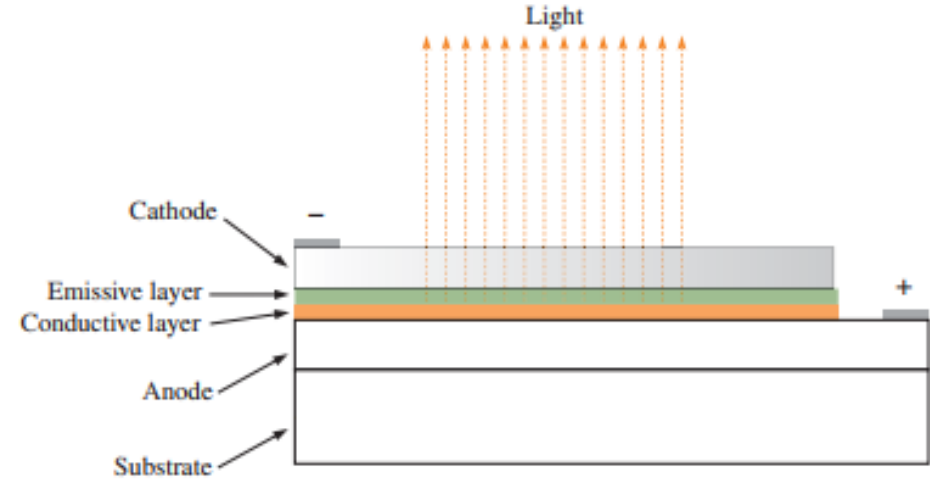


Fig 3.42 — Basic structure of a 2-layer OLED

The Photodiode — Light-Controlled Current

• Light In, Current Out

- A photodiode operates in reverse bias, with a window that lets light reach the junction
- The tiny current that is present when the diode is not exposed to light is called dark current

• The Key Behaviour

- Shine light on it and the reverse current rises with the light intensity (irradiance, in mW/cm^2)
- So it behaves as a light-controlled variable resistance — the opposite role of an LED



Fig 3.44 — The photodiode — operation, devices, and symbol

Photodiode Operation — Dark vs. Illuminated

- **No Light → Dark Current**

- Reverse-biased with no incident light, only a very small dark current flows

- **Light → Proportional Reverse Current**

- When a light beam strikes the junction, the reverse current rises in proportion to the light intensity (irradiance)
- This is exactly what lets the photodiode act as a light-controlled current source — or, as the previous example showed, a light-controlled variable resistance

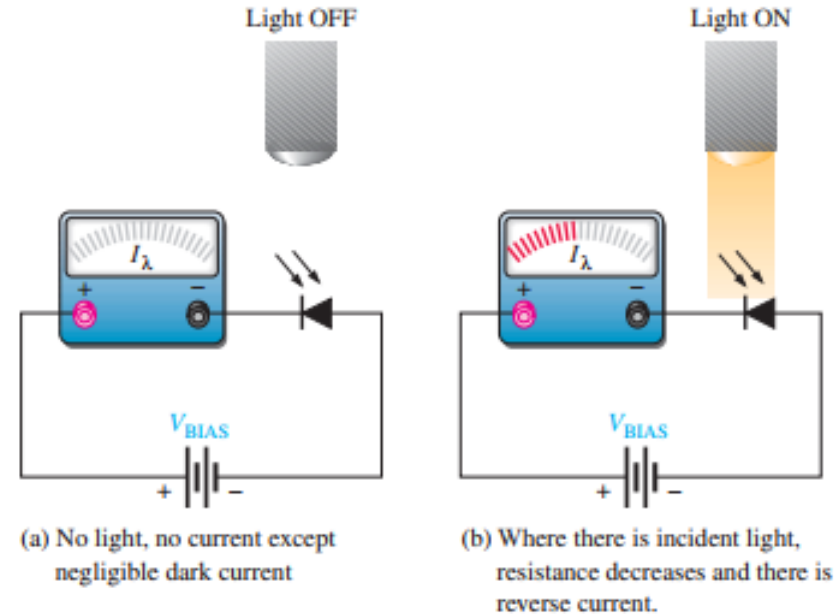


Fig 3.46 — Operation of a photodiode

Worked Example — Photodiode as a Variable Resistance

• The Question

- At $V_R = 10\text{ V}$, a photodiode passes $1.4\text{ }\mu\text{A}$ in dim light (0.5 mW/cm^2) and $55\text{ }\mu\text{A}$ in bright light (20 mW/cm^2). Find its resistance in each case.

• The Solution

- Dim

$$R_R = \frac{V_R}{I_A} = \frac{10\text{ V}}{1.4\text{ }\mu\text{A}} = 7.14\text{ M}\Omega$$

- Bright

$$R_R = \frac{V_R}{I_A} = \frac{10\text{ V}}{55\text{ }\mu\text{A}} = 182\text{ k}\Omega$$

• The Point

- More light $\rightarrow$ more current $\rightarrow$ lower resistance: the photodiode is a light-controlled resistor

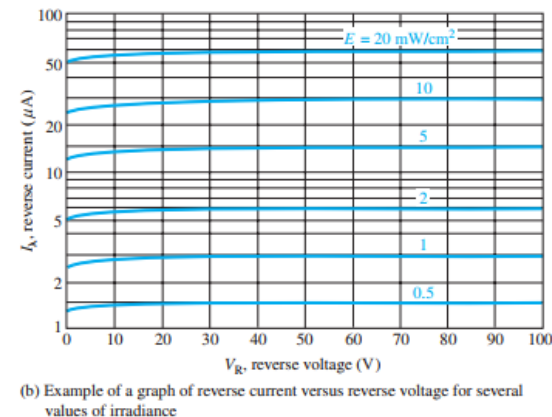
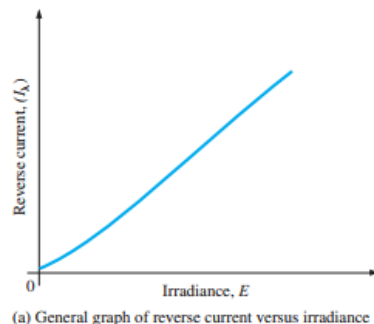


Fig 3.45 — Typical photodiode characteristics

Quick Check #2 — Optical Diodes

- **Question 1**

- What process produces light in an LED, and what sets the colour?

- **Question 2**

- Why does every LED need a series current-limiting resistor?

- **Question 3**

- Give two advantages of high-intensity LEDs over incandescent bulbs.

- **Question 4**

- How do quantum dots help produce a more natural white light?

- **Question 5**

- In what bias does a photodiode operate, and how does its current respond to light?

Part 3 — Other Special-Purpose Diodes

The Schottky Diode

- **A Metal–Semiconductor Junction**

- High-current diodes used primarily in high-frequency and fast-switching applications
- Formed by joining a (usually n-type) semiconductor to a metal — not a pn junction
- Forward voltage is low, typically about 0.3 V (there's no normal depletion region)

- **Why It Matters**

- It conducts with majority carriers only — no minority carriers, so almost no reverse-recovery delay
- A very fast, high-frequency switching diode



Fig 3.49 — Schottky diode symbol

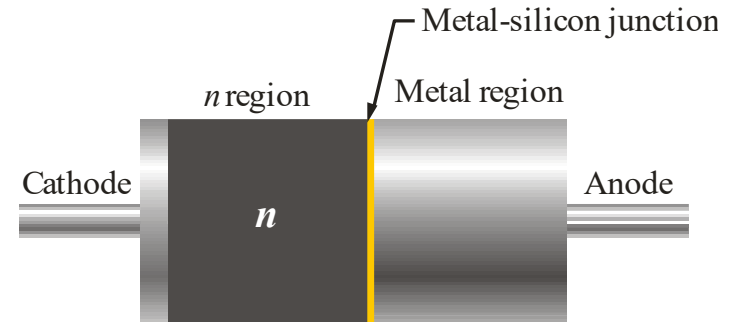


Fig 3.50 — Internal construction of a Schottky diode

The PIN Diode

- **Three Regions: P – I – N**

- Heavily doped p and n regions separated by an intrinsic (undoped) region

- **Two Modes**

- Reverse-biased: behaves like a nearly constant capacitance
- Forward-biased: behaves like a current-controlled variable resistance (resistance falls as current rises)
- Used as a microwave switch, a modulator, and a radio frequency (RF) attenuator; some serve as fiber-optic photodetectors

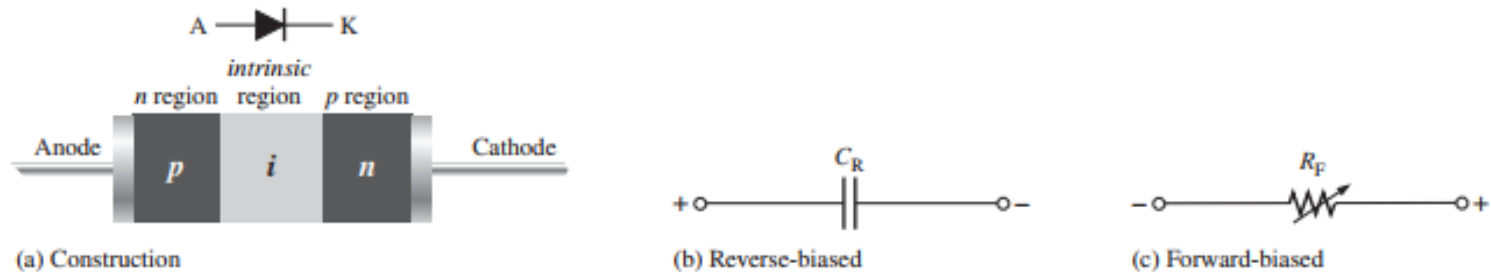


Fig 3.51 — The PIN diode — construction and bias behaviour

The Tunnel Diode

- **Negative Resistance**

- Very heavily doped → an extremely thin depletion region that lets electrons 'tunnel' across at low forward voltage
- Over part of its curve, current DECREASES as voltage increases — a negative-resistance region

- **Application — Oscillators**

- Biased in that region, it cancels the losses of a resonant tank circuit, sustaining oscillation
- Used in oscillators and microwave amplifiers at very high frequencies

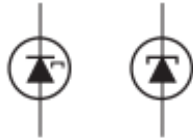


Fig 3.53— Tunnel diode symbols

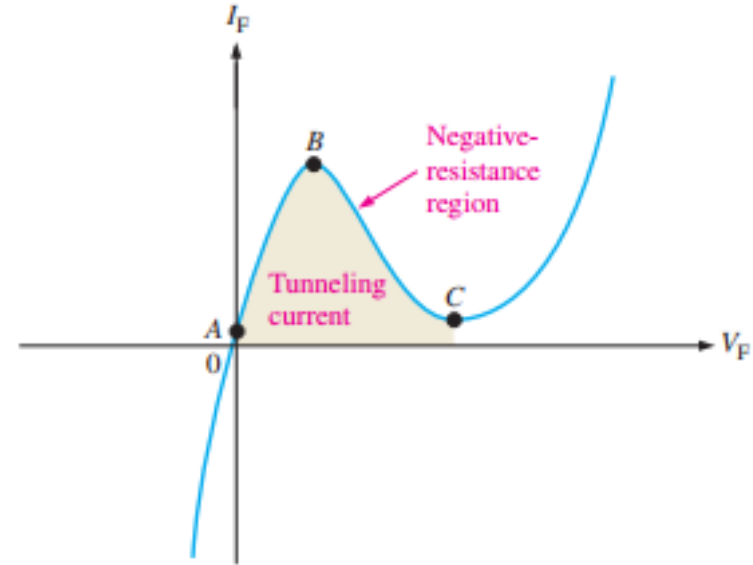


Fig 3.54 — Tunnel-diode characteristic curve (negative-resistance region)

The Laser Diode

- **Coherent Light by Stimulated Emission**

- LASER = Light Amplification by Stimulated Emission of Radiation
- Forward-biased like an LED, but built with a resonant cavity (reflective ends) so emitted photons trigger more identical photons

- **LED vs. Laser**

- Above a threshold current it emits coherent, single-wavelength light (an LED emits incoherent light)
- Used in CD/DVD pickups (with a photodiode), laser printers, and fiber-optic systems

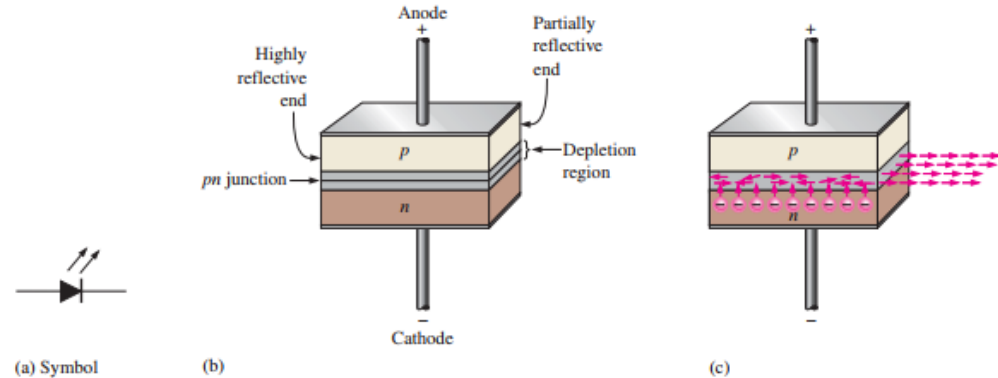


Fig 3.48 — Basic laser-diode construction and operation

The Current-Regulator Diode

- **Constant Current — the Zener's Mirror Image**

- Where a zener holds voltage constant, the current-regulator (constant-current) diode holds CURRENT constant
- It operates in forward bias; the current settles at the regulator value I_P

- **Operating Range**

- Regulation runs from the limiting voltage V_L up to the peak operating voltage (POV)
- Between those, the forward current stays essentially constant regardless of voltage

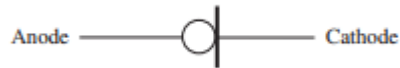


Fig 3.57 — Symbol for current regulator diode

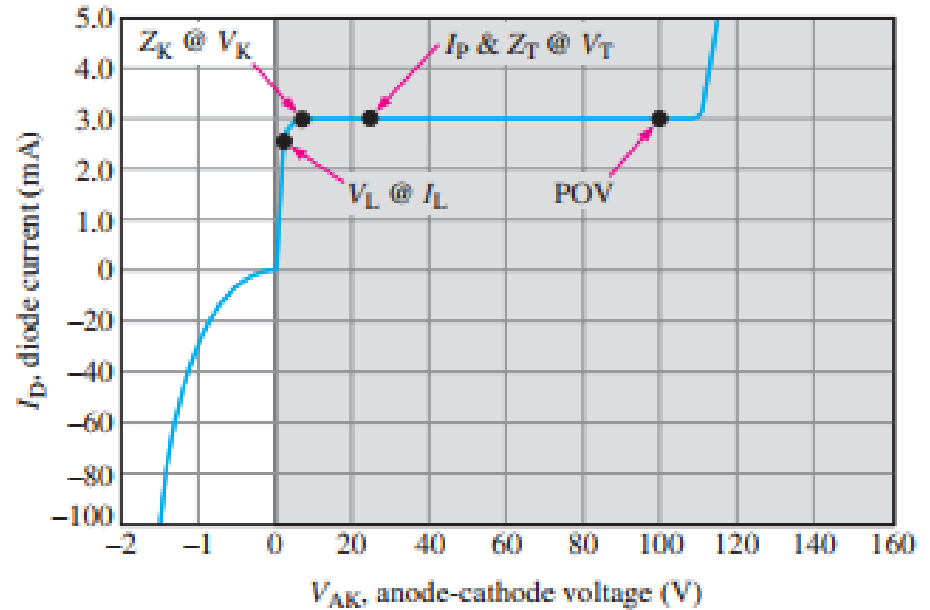


Fig 3.58 — Current-regulator-diode characteristic curve

Quick Check #3 — Other Special Diodes

- **Question 1**

- What gives a Schottky diode its low forward voltage and fast switching?

- **Question 2**

- How does a PIN diode behave under reverse versus forward bias?

- **Question 3**

- What is the defining characteristic of a tunnel diode, and one use?

- **Question 4**

- How does a laser diode's light differ from an LED's?

- **Question 5**

- Contrast the step-recovery diode with the current-regulator diode.

Part 4 — Wrap-Up

- **Each Built Around One Property**

- Varactor — voltage-controlled capacitance → tuning
- LED — emits light when forward-biased → indicators, displays, lighting
- Photodiode — light controls reverse current → light sensing
- Schottky — metal junction, ~ 0.3 V, majority-only → fast, high-frequency switching
- PIN — capacitance (reverse) / variable resistance (forward) → microwave switching & attenuation
- Tunnel — negative resistance → high-frequency oscillators; Laser — coherent light → optical storage, fiber
- Current-regulator — holds constant current

Key Takeaways from Today

- Special diodes each exploit one property of the junction beyond simple rectification
- Varactor: a reverse-biased diode as a voltage-controlled capacitor (C falls as reverse voltage rises) — for tuning
- LED: forward bias $\rightarrow$ electroluminescence; colour set by band gap; $V_F \approx 1.2\text{--}3.2\text{ V}$; always use a current-limiting resistor
- LED family: high-intensity LEDs for lighting/signals; OLEDs (thin, flexible); quantum dots (size sets colour)
- Photodiode: reverse bias $\rightarrow$ reverse current rises with light intensity; a light-controlled resistance
- Schottky ($\sim 0.3\text{ V}$, fast), PIN (RF switch/attenuator), tunnel (negative resistance), laser (coherent light)
- Current-regulator: constant current — the zener's mirror image

Coming Up — Week 8: BJT Operation & Characteristics

- **Next Topic — The Bipolar Junction Transistor (Lecture Notes pp. 117–135)**
 - From two-terminal diodes to a three-terminal device — the transistor
 - BJT structure (npn and pnp), how it's biased, and transistor action
- **To Prepare**
 - You'll combine two pn junctions back-to-back — keep forward and reverse bias clear in mind
- **Reminder**
 - Read pp. 117–135 before class

Thank You — Questions?